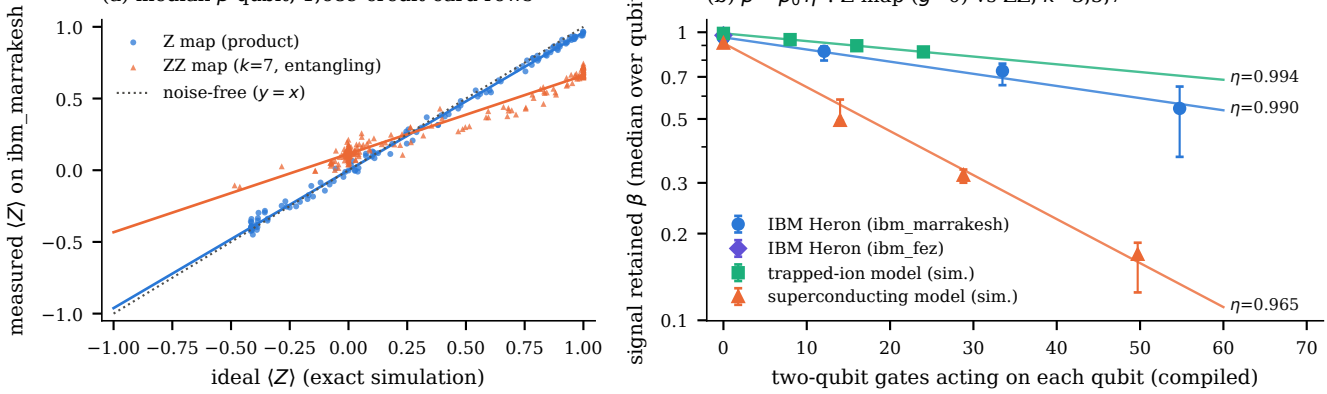
(a) median-β qubit, 1,689 credit-card rows
Z map (product)
ZZ map (k=7, entangling)
noise-free (y = x)
measured ⟨Z⟩ on ibm_marrakesh
ideal ⟨Z⟩ (exact simulation)
1.0
0.5
0.0
−0.5
−1.0
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
(b) β ≈ β0 η^g: Z map (g=0) vs ZZ, k=3,5,7
signal retained β (median over qubits)
two-qubit gates acting on each qubit (compiled)
η=0.994
η=0.990
η=0.965
IBM Heron (ibm_marrakesh)
IBM Heron (ibm_fez)
trapped-ion model (sim.)
superconducting model (sim.)
1
0.7
0.5
0.3
0.2
0.1
0
10
20
30
40
50
60
70